

Caroline Piaulet-Ghorayeb

MARGARET BURBRIDGE PRIZE POSTDOCTORAL FELLOW · UNIVERSITY OF CHICAGO

✉ carolinepiaulet@uchicago.edu | 🏠 cpiaulet.github.io | 📄 github.com/cpiaulet

Publications

6 1ST-AUTHOR, 51 REF'D (+11 SUBM.), 2,600+ CITATIONS, H-INDEX=27, ORCID: 0000-0002-2875-917X

[Legend: * means not peer-reviewed or ongoing review process]

- [***64**] **Piaulet-Ghorayeb** et al., “A window for water-hydrogen demixing on metal-rich warm sub-Neptunes” (2025, submitted to ApJ)
- [***63**] Kubyshkina, Egger & **Piaulet-Ghorayeb**, “Escape of Water- and Metal-enriched Atmospheres from compact Hot mini-Neptunes with CHAIN” (2025, submitted to A&A)
- [***62**] Lamontagne, P., et al. (incl. **Piaulet-Ghorayeb**), “NIRPS tightens the mass estimate of GJ 3090 b and detects a planet near the stellar rotation period” (2025, submitted to A&A)
- [***61**] Frensch, Y., et al. (incl. **Piaulet-Ghorayeb**), “TOI-3288 b and TOI-4666 b: two gas giants transiting low-mass stars characterised by NIRPS” (2025, submitted to A&A)
- [***60**] Splinter, J., et al. (incl. **Piaulet-Ghorayeb**), “Precise Constraints on the Energy Budget of WASP-121 b from its JWST NIRISS/SOSS Phase Curve” (2025, submitted to AAS Journals, arXiv:2509.09760)
- [***59**] **Piaulet-Ghorayeb**, “STCTM: a forward modeling and retrieval framework for stellar contamination and stellar spectra” (2025, submitted to JOSS, arXiv:2508.19297)
- [***58**] Pelletier, S., et al. (incl. **Piaulet-Ghorayeb**), “Enriched volatiles and refractories but deficient titanium on the dayside atmosphere of WASP-121b revealed by JWST/NIRISS” (2025, submitted to A&A, arXiv:2508.18341)
- [***57**] Xue, Q., et al. (incl. **Piaulet-Ghorayeb**), “The JWST Rocky Worlds DDT Program reveals GJ 3929b to likely be a bare rock” (2025, submitted to AAS Journals, arXiv:2508.12516)
- [***56**] Feinstein, A., et al. (incl. **Piaulet-Ghorayeb**), “On Linking Planet Formation Models, Protoplanetary Disk Properties, and Mature Gas Giant Exoplanet Atmospheres” (2025, submitted to AAS Journals, arXiv:2506.00669)
- [***55**] Crossfield, I., et al. (incl. **Piaulet-Ghorayeb**), “Mapping the SO₂ shoreline in gas giant exoplanets” (2025, submitted to AAS Journals, arXiv:2509.14318)
- [***54**] Benneke, B., Roy, P.-A., **Piaulet, C.**, et al., “JWST NIRSpec Reconnaissance Transmission Spectroscopy of the Habitable-zone Exo-Earth TRAPPIST-1g” (2024, submitted to **Nature**)
- [**53**] Allart, R., et al. (incl. **Piaulet-Ghorayeb**), “A complex structure of escaping helium spanning more than half the orbit of the ultra-hot Jupiter WASP-121 b” (2025, accepted in **Nature Communications**)
- [**52**] Parc, L., et al. (incl. **Piaulet-Ghorayeb**), “NIRPS and TESS reveal a peculiar system around the M dwarf TOI-756: A transiting sub-Neptune and a cold eccentric giant” (2025, accepted in **A&A**)
- [**51**] Brady, M., et al. (incl. **Piaulet-Ghorayeb**), “An Earth-like Density for the Temperate Earth-sized Planet GJ 12b” (2025, accepted for publication in AJ, arXiv:2506.20561)
- [**50**] Coulombe, L.-P., Benneke, B., Krissansen-Totton, J., L’Heureux, A., **Piaulet-Ghorayeb, C.**, et al., “Tentative evidence for the presence of volatiles on the warm super-Earth TOI-270 b” (2025, AJ, 170, 4, 226)
- [**49**] Bazinet, L., et al. (incl. **Piaulet-Ghorayeb**), “Quantifying thermal water dissociation in the dayside photosphere of WASP-121 b using NIRPS” (2025, accepted in A&A, arXiv:2508.06626)
- [**48**] Krishnamurthy, V., Carteret, Y., **Piaulet-Ghorayeb**, et al., “Continuous helium absorption from the leading and trailing tails of WASP-107b” (2025, accepted to **Nature Astronomy**, arXiv: 2505.20588)
- [**47**] Cadieux, C., L’Heureux, A., **Piaulet-Ghorayeb**, et al., “Detailed Architecture of the L 98-59 System and Confirmation of a Fifth Planet in the Habitable Zone” (2025, AJ, 170, 154)
- [**46**] **Piaulet-Ghorayeb**, et al., “Strict Limits on Potential Secondary Atmospheres on the Temperate Rocky Exo-Earth TRAPPIST-1 d” (2025, ApJ, 989, 181)
- [**45**] Luque, R., **Piaulet-Ghorayeb**, et al., “Insufficient evidence for DMS and DMDS in the atmosphere of K2-18 b: From a joint analysis of JWST NIRISS, NIRSpec, and MIRI observations” (2025, A&A, 700, A284)

- [44] Gomes da Silva, J., **et al. (incl. Piaulet-Ghorayeb)**, “Blind search for activity-sensitive lines in the near-infrared using HARPS and NIRPS observations of Proxima and Gl 581” (2025, A&A, 700, A177)
- [43] Suarez Mascareño, A., **et al. (incl. Piaulet-Ghorayeb)**, “Diving into the planetary system of Proxima with NIRPS: Breaking the metre per second barrier in the infrared” (2025, A&A, 700, A11)
- [42] Bouchy, F., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS joining HARPS at ESO 3.6 m: On-sky performance and science objectives” (2025, A&A, 700, A10)
- [41] Vulato, V., **et al. (incl. Piaulet-Ghorayeb)**, “Hydride ion continuum hides absorption signatures in the NIRPS near-infrared transmission spectrum of the ultra-hot gas giant WASP-189b” (2025, A&A, 700, A9)
- [40] Allart, R., **et al. (incl. Piaulet-Ghorayeb)**, “NIRPS detection of delayed atmospheric escape from the warm and misaligned Saturn-mass exoplanet WASP-69 b” (2025, A&A, 700, A7)
- [39] Taylor, J., **et al. (incl. Piaulet-Ghorayeb)**, “JWST NIRISS transmission spectroscopy of the super-Earth GJ 357b, a favourable target for atmospheric retention” (2025, MNRAS, 540, pp. 3677-3692)
- [38] Fournier-Tondreau, M., **et al. (incl. Piaulet-Ghorayeb)**, “Transmission spectroscopy of WASP-52 b with JWST NIRISS: water and helium atmospheric absorption, alongside prominent star-spot crossings” (2025, MNRAS, 539, pp. 422-438)
- [37] Ahrer, E.-M., Radica, M., **Piaulet-Ghorayeb**, et al., “Escaping Helium and a Highly Muted Spectrum Suggest a Metal-enriched Atmosphere on Sub-Neptune GJ 3090 b from JWST Transit Spectroscopy” (2025, ApJL, 985, L10)
- [36] Morel, K., **et al. (incl. Piaulet-Ghorayeb)**, “A Moderate Albedo from Reflecting Aerosols on the Dayside of WASP-80 b Revealed by JWST/NIRISS Eclipse Spectroscopy” (2025, AJ, 169, 277)
- [35] Monaghan, C., **et al. (incl. Piaulet-Ghorayeb)**, “Low 4.5 μm Dayside Emission Disfavors a Dark Bare-rock Scenario for the Hot Super-Earth TOI-431 b” (2025, AJ, 169, 239)
- [34] Radica, M., **Piaulet-Ghorayeb, C.**, et al., “Promise and Peril: Stellar Contamination and Strict Limits on the Atmosphere Composition of TRAPPIST-1c from JWST NIRISS Transmission Spectra” (2025, ApJL, 979, L5)
- [32] **Piaulet-Ghorayeb, C.**, et al., “JWST/NIRISS reveals the water-rich “steam world” atmosphere of GJ 9827 d” (2024, ApJL, 974, L10)
- [31] Bell, T., **et al. (Piaulet: Tier 3)**, “Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b” (2023, **Nature Astro.**, 8, 879-898, Volume 8, p. 879-898)
- [30] **JWST TRAPPIST-1 Community Initiative (incl. Piaulet)**, “A roadmap to the efficient and robust characterization of temperate terrestrial planet atmospheres with JWST” (2024, **Nature Astro.**, 8, 810-818)
- [29] Cadieux, C. **et al. (incl. Piaulet-Ghorayeb)**, “Transmission Spectroscopy of the Habitable Zone Exoplanet LHS 1140 b with JWST/NIRISS” (2024, ApJL, 970, L2)
- [*28] Artigau, E. and **NIRPS Consortium (incl. Piaulet)**, “NIRPS first light and early science: breaking the 1 m/s RV precision barrier at infrared wavelengths” (2024, SPIE Proceedings, arXiv:2406.08304)
- [*27] Benneke, B., Roy, P.-A., Coulombe, L.-P., Radica, M., **Piaulet, C.**, et al., “JWST Reveals CH₄, CO₂, and H₂O in a Metal-rich Miscible Atmosphere on a Two-Earth-Radius Exoplanet” (2024, submitted to ApJL, arXiv: 2403.03325)
- [26] Fournier-Tondreau, M., MacDonald, R.J., Radica, M., Lafrenière, D., Welbanks, L., **Piaulet, C.**, et al., “Near-Infrared Transmission Spectroscopy of HAT-P-18 b with NIRISS: Disentangling Planetary and Stellar Features in the Era of JWST” (2024, MNRAS, 528, 3354)
- [25] Howard, W. S., **et al. (incl. Piaulet)**, “Characterizing the Near-Infrared Spectra of flares from TRAPPIST-1 during JWST Transit Spectroscopy Observations” (2023, ApJ, 959, 64)
- [24] Lim, O., Benneke, B., Doyon, R., Macdonald, R. J., **Piaulet, C.**, et al., “Atmospheric Reconnaissance of TRAPPIST-1 b with JWST/NIRISS: Evidence for Strong Stellar Contamination in the Transmission Spectra” (2023, ApJL, 955, L22)
- [23] Roy, P.-A., Benneke, B., **Piaulet, C.**, et al., “Water absorption in the transmission spectrum of the water-world candidate GJ 9827 d” (2023, ApJL, 954, L52)
- [22] Coulombe, L.-P., **et al. (Piaulet: Tier 3)**, “A broadband thermal emission spectrum of the ultra-hot Jupiter WASP-18b” (2023, **Nature**, 620, 292)
- [21] Albert, L., **et al. (incl. Piaulet)**, “The Near Infrared Imager and Slitless Spectrograph for the James Webb Space Telescope – III. Single Object Slitless Spectroscopy” (2023, PASP, 135, 075001)

- [20] Peterson, M., Benneke, B., Collins, K., **Piaulet, C.**, et al., “A temperate Earth-sized planet with tidal heating transiting an M6 star” (2023, **Nature**, 617, 701) **Note: Benneke & Piaulet lead authors after M. Peterson left the field in 2021.**
- [19] Feinstein, A., **et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with NIRISS-SOSS” (2023, **Nature** 614, 670–675)
- [18] Ahrer, E.-M., **et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRCам” (2023, **Nature** 614, 653–658)
- [17] Alderson, L., **et al. (Piaulet: Tier 3)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H” (2023, **Nature** 614, 664–669)
- [16] Rustamkulov, Z., **et al. (Piaulet: Top Tier 1/4)**, “Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM” (2023, **Nature** 614, 659–663)
- [15] **The JWST Transiting Exoplanet Community Early Release Science Team**, “Identification of carbon dioxide in an exoplanet atmosphere” (2023, **Nature** 614, 649–652)
- [14] **Piaulet, C.**, et al. “Evidence for the volatile-rich composition of a 1.5-Earth-radius planet” (2023, **Nature Astro.** 7, 206–222)
- [13] Roy, P.-A., Benneke, B., **Piaulet, C.**, et al., “Is the Hot, Dense Sub-Neptune TOI-824 b an Exposed Neptune Mantle? Spitzer Detection of the Hot Dayside and Reanalysis of the Interior Composition” (2022, *ApJ*, 941, 89)
- [12] Bell, T., **et al. (incl. Piaulet: Tier 2)**, “Eureka!: An End-to-End Pipeline for JWST Time-Series Observations” (2022, *JOSS*, 7, 4503)
- [11] Radica, M., **et al. (incl. Piaulet)**, “APPLESOSS: A Producer of Profiles for SOSS. Application to the NIRISS SOSS Mode” (2022, *PASP*, 134, 104502)
- [10] Darveau-Bernier, A., **et al. (incl. Piaulet)**, “ATOCA: an Algorithm to Treat Order Contamination. Application to the NIRISS SOSS Mode” (2022, *PASP*, 134, 094502)
- [9] Morris, B. M., Heng, K., Jones, K., **Piaulet, C.**, et al., “Physically-motivated basis functions for temperature maps of exoplanets” (2022, *A&A*, 660, A123)
- [8] Zeng, L., Jacobsen, S. B., Hyung, E., Levi, A., Nava, C., Kirk, J., **Piaulet, C.**, et al., “New Perspectives on the Exoplanet Radius Gap from a Mathematica Tool and Visualized Water Equation of State” (2021, *ApJ*, 923, 247)
- [7] Pelletier, S., Benneke, B., Darveau-Bernier, A., Boucher, A., Cook, N. J., **Piaulet, C.**, et al., “Where Is the Water? Jupiter-like C/H Ratio but Strong H₂O Depletion Found on τ Boötis b Using SPIRou” (2021, *AJ*, 162, 73)
- [6] **Piaulet, C.**, et al., “WASP-107b’s density is even lower: a case study for the physics of planetary gas envelope accretion and orbital migration” (2021, *AJ*, 161, 70)
- [5] Kosiarek, M., Berardo, David A., Crossfield, I. J. M., Laguna, C., **Piaulet, C.**, et al., “Physical Parameters of the Multi-Planet Systems HD 106315 and GJ 9827” (2021, *AJ*, 161, 47)
- [4] St-Louis, L., **Piaulet, C.**, et al., “An extensive spectroscopic time series of three Wolf-Rayet stars - II. A search for wind asymmetries in the dust-forming WC7 binary WR137” (2020, *MNRAS*, 497, 4448)
- [3] Benneke, B., Wong, I., **Piaulet, C.**, et al., “Water Vapor and Clouds on the Habitable-zone Sub-Neptune Exoplanet K2-18b” (2019, *ApJL*, 887, L14)
- [2] Ramiamananantsoa, T. **et al. (incl. Piaulet)**, “BRITE-Constellation high-precision time-dependent photometry of the early-O-type supergiant ζ Puppis unveils the photospheric drivers of its small- and large-scale wind structures” (2018, *MNRAS*, 473, 5532)
- [*1] **Piaulet, C.**, Mochizuki, K., Arguin, J.-F., “Improvement of the multivariate analysis method for the reduction of electron charge misidentification in ATLAS data” (2017, ATL-COM-PHYS-2017-1539)